

When do we need to center the Fourier transform of an image? Prove that multiplying the image intensities by $(-1)^{x+y}$ where (x,y) represents the coordinate of pixels, centers the Fourier transform?

Solution:

2D Discrete Fourier Transform places the DC components (average intensity) at the top left corner $(0,0)$ of the frequency domain image. However for better visualization and processing, DC component need to appear at the center of spectrum.

Reasons to be DC spectrum at center:

- 1) For symmetry in visual representation
- 2) When applying frequency domain filters that assume the center is the DC component.

Proof: How multiplying by $(-1)^{x+y}$ centers the DFT.

Step 1: We have 2D DFT of image $f(x,y)$ of size $M \times N$ is:

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

Step 2: Apply centering in spatial domain

From Euler's identity

$$e^{i\theta} = \cos\theta + j\sin\theta$$

substitute $\theta = \pi(x+y)$

$$e^{j\pi(x+y)} = \cos(\pi(x+y)) + j\sin(\pi(x+y))$$

We know, $\cos n\pi = (-1)^n$ for any integer n
 $\sin n\pi = 0$ for any integer n .

Since $x+y$ is an integer,

$$\begin{aligned} e^{j\pi(x+y)} &= \cos(\pi(x+y)) + j \cdot 0 \\ &= (-1)^{x+y} \end{aligned}$$

Multiply $f(x,y)$ by $(-1)^{x+y} = e^{j\pi(x+y)}$

$$f_c(x,y) = f(x,y) \cdot e^{j\pi(x+y)}$$

Now take the DFT of $f_c(x,y)$

$$F_c(x,y) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) \cdot e^{j\pi(x+y)} \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

$$F_c(x,y) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N} - \frac{x}{2} - \frac{y}{2}\right)}$$

$$F_c(x,y) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi\left(\frac{(u-M/2)x}{M} + \frac{(v-N/2)y}{N}\right)}$$

which is

$$F_c(u,v) = F\left(u - \frac{M}{2}, v - \frac{N}{2}\right)$$

so the spectrum is shifted by $\left(\frac{M}{2}, \frac{N}{2}\right)$

The DC component of $F(0,0)$ moves to $\left(\frac{M}{2}, \frac{N}{2}\right)$ i.e the center of the image.